A Morphological-Gradient Characterisation of Dead-Tree Clusters in Southeastern Finland's Boreal Forests

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Abstract

Finland's boreal forests are experiencing a significant and documented increase in non-harvest tree mortality [1]. Characterising the spatial morphology of mortality clusters is a prerequisite for hypothesising the disturbance processes that generate them. We present a regional-scale analysis of 698,223 dead-tree locations identified from aerial imagery in southeastern Finland in 2023. Using DBSCAN (eps = 20 m, min_samples = 3) we derived 14,582 quality-filtered clusters and characterised each with a suite of spatial metrics including the Donnelly-corrected Clark–Evans index, fractal dimension, compactness, aspect ratio, alpha compactness, core-to-edge ratio, and within-cluster nearest-neighbour distance. K-means clustering on three morphological features ($k = 2$; silhouette = 0.401) recovers a dominant *compactness–elongation gradient* that exceeds a covariance-preserving null (single multivariate Gaussian matched to the observed feature covariance; $p = 0.001$): the gradient itself is real. Its *discrete* two-type reading, however, is not. We subject the partition to four stability gates, re-clustering from the raw points: (i) it is not invariant to the DBSCAN radius (endpoint prevalence drifts 22.9–28.6% across eps 10–40 m, with continuously migrating centroids); (ii) its 23.5 / 76.5% split is an artefact of the aspect-ratio filter, dissolving into a single continuous tail once the cap is released; (iii) the elongated endpoint carries *no* directional (anisotropic) signal beyond hull shape—and therefore no wind-throw fingerprint—when matched on aspect ratio; and (iv) type identity is partly predictable from detector confidence alone (AUC = 0.727). Mixture-model BIC and a unimodal principal axis independently favour a continuum over two modes, and a held-out Random Forest that appears to recover the split (AUC = 0.976 ± 0.002) does so only through geometric redundancy among shape features—the one shape-orthogonal feature, landscape connectivity, adds nothing. We therefore report a morphological gradient, not a typology, and withhold disturbance-agent labels: with no species, dating, or ground-truth layers—and the only public agent maps fusing wind and beetle into a single class—attribution is not supportable. The central contribution is thus partly cautionary: discrete, agent-labelled typologies derived from DBSCAN-then-shape pipelines can be artefacts of clustering scale, filter thresholds, and detector behaviour, and should be tested against these gates before being read as ecology.

Keywords: boreal forest mortality; spatial clustering; DBSCAN; Clark–Evans index; morphological gradient; cluster shape metrics; clustering-scale sensitivity; Finland

1 Introduction

Boreal forests store roughly 30% of terrestrial carbon and support exceptional biodiversity, yet they are undergoing accelerating disturbance as climate change intensifies drought, windstorms, and pest outbreaks [2; 3]. In Finland, non-harvest tree mortality has increased substantially in recent years [1], a trajectory that demands systematic monitoring. Understanding which ecological processes drive this mortality—and where they concentrate across the landscape—is a prerequisite for adaptive management and carbon-flux modelling.

The spatial arrangement of dead trees encodes information about the process that killed them. Bark-beetle outbreaks typically generate dense, spatially contagious patches as infestation spreads from tree to tree [4; 5]. Wind damage, by contrast, tends to produce elongated, directionally oriented swaths aligned with storm tracks [6; 7]. Drought-induced mortality can produce more scattered, diffuse patterns tied to soil-water gradients [8]. These morphological *fingerprints* are in principle recoverable from point-cloud data even when individual tree species and cause-of-death records are unavailable.

High-resolution aerial imagery now enables the detection and georeferencing of individual dead trees over large areas. The 2023 survey of southeastern Finland used here produced 698,223 dead-tree locations in ETRS89/TM35FIN (EPSG:3067) coordinates—a dataset large enough to support robust statistical characterisation of cluster morphology at scales ranging from the individual stand to the regional landscape.

This paper makes three methodological contributions:

1. **A reproducible large-scale pipeline with the Donnelly-corrected Clark–Evans index.** We apply DBSCAN to the full 698,223-point dataset and compute the Donnelly (1978) edge-corrected Clark–Evans statistic for each cluster, enabling unbiased comparison of internal point dispersion across clusters of varying size and shape.

2. **A morphological compactness–elongation gradient, tested for discreteness.** K-means clustering on fractal dimension, Clark–Evans index, and aspect ratio recovers a dominant one-dimensional gradient. We show this structure exceeds a *covariance-preserving* null (a single multivariate Gaussian matched to the observed feature covariance; $p = 0.001$)—a stricter test than column-shuffling, which destroys all inter-feature correlation. We further show, via mixture-model BIC and a unimodal primary axis, that the gradient is *not* a clean two-way dichotomy; the $k = 2$ split is offered as an interpretive summary, not a discovered class boundary.

3. **Stability gates that separate a robust gradient from a fragile partition.** Re-clustering from raw, we show the continuous gradient is stable but the *discrete* two-class reading is not: the split is not invariant to the DBSCAN radius, its prevalence is an artefact of the aspect-ratio filter, it carries no directional structure beyond hull shape, and it is partly predictable from detector confidence alone. A held-out Random Forest reproduces the $k = 2$ partition (AUC $= 0.976$) only through geometric redundancy among shape descriptors—the one shape-orthogonal feature (landscape connectivity) adds no signal. We therefore present a morphological gradient, explicitly not a disturbance typology.

Throughout, we deliberately withhold disturbance-agent labels. Morphological associations with wind, bark beetle, or other agents are testable hypotheses, not findings: confirmed attribution would require species records, mortality dates linked to known events (the available `alku_aika` field records flight-strip acquisition time, not mortality date), and multi-temporal imagery, none of which are present in the current dataset. We return to specific attribution caveats—including the absence of any directional signal in the elongated type—in the Discussion.

2 Related Work

European forest disturbance ecology and mapping. Europe's forests have been subject to intensifying disturbance regimes over recent decades, driven by the interaction of climate change with biotic and abiotic agents [9; 10]. Senf and Seidl [11] produced the first continent-wide map of forest disturbance regimes using satellite time series, revealing strong latitudinal gradients in the relative importance of wind, bark beetles, and fire. Allen et al. [3] documented globally the accelerating role of drought and heat as pre-disposing stressors that increase forest vulnerability to secondary agents. These landscape-scale perspectives highlight the need for methods that can characterise disturbance not just by area affected, but by the spatial structure of affected patches.

Bark beetle ecology and spatial signatures. The Eurasian spruce bark beetle (*Ips typographus*) is the dominant biotic disturbance agent in boreal and subalpine Europe [5]. Raffa et al. [4] showed that bark-beetle eruptions are governed by cross-scale feedbacks: fine-scale tree-to-tree contagion at the stand level, mediated by pheromone plumes and resource depletion, scales up to landscape-level epidemics when predisposing conditions (drought, windthrow) are met. Jönsson et al. [12] analysed epidemic dynamics of *Ips typographus* in Sweden, identifying that outbreaks initiated in windthrown timber and then spread as self-reinforcing patch growth. Kautz et al. [13] demonstrated that high-resolution empirical host-tree distributions can be used to simulate outbreak spread, predicting the compact, dense patch morphology that characterises beetle kills. Wichmann and Ravn [14] documented how storm-felled timber from the 1999 Danish windstorm acted as a beetle breeding reservoir, amplifying subsequent outbreaks and leaving spatially contagious kill patches.

Wind damage in Fennoscandia. Wind is the leading abiotic disturbance agent in northern European forests [15]. Gardiner et al. [6] reviewed how silvicultural history, stand structure, and edge exposure interact to determine windstorm susceptibility, noting that clearcut edges and monoculture Norway spruce stands are particularly vulnerable. Gregow et al. [7] specifically modelled combined wind and snow-loading risks in Finland under current and projected future climate, finding that snow-induced stem damage concentrated in central and eastern Finland may increase with warming winters. The elongated, directional morphology of wind-felled areas has been noted in several Fennoscandian studies as a diagnostic feature distinguishing wind damage from beetle-induced mortality.

Spatial statistics methods. The Clark–Evans statistic [16] remains the canonical index of point dispersion within bounded study regions. Donnelly [17] derived an analytical edge correction that removes the downward bias introduced by boundary effects in finite study areas, making the corrected statistic suitable for small clusters with irregular convex-hull boundaries. Anselin [18] introduced Local Indicators of Spatial Association (LISA), enabling the identification of statistically significant spatial clusters and outliers at multiple scales. Wiegand and Moloney [19] provide a comprehensive treatment of spatial point-pattern analysis methods for ecological applications, including the mark correlation function, Ripley's K, and related second-order statistics that complement the first-order metrics used here.

3 Methods

3.1 Dataset

The dataset comprises 698,223 dead-tree locations predicted from a deep-learning model applied to high-resolution aerial imagery acquired in 2023 over south-eastern Finland [1]. Coordinates are provided in ETRS89/TM35FIN (EPSG:3067), a conformal transverse Mercator projection with metric units, appropriate for distance-based spatial analyses. The dataset covers a regional footprint, not the entire country; reported cluster prevalences and densities are therefore regional and should not be read as national statistics.

3.2 Clustering

Density-based spatial clustering was performed using DBSCAN [20] with neighbourhood radius eps = 20 m and minimum neighbourhood size min_samples = 3. The eps = 20 m threshold was chosen to capture within-stand tree-to-tree spread typical of both bark-beetle and wind disturbance, while separating clusters arising from distinct events. The 43,539 raw clusters were post-filtered to retain only those satisfying:

- $n \geq 5$ trees (excludes trivially small groups),
- area $\geq 100\,\mathrm{m}^2$ (minimum ecologically meaningful patch size),
- aspect ratio ≤ 10 (removes linear artefacts such as road corridors).

This yielded 14,582 clusters for analysis.

3.3 Spatial Metrics

Each cluster was characterised by the following spatial metrics.

Within-cluster nearest-neighbour distance (NND). For a cluster containing n points, the NND for point i is

$$d_i = \min_{j \neq i} \|\mathbf{p}_i - \mathbf{p}_j\|, \tag{1}$$

and the cluster-level summary statistics are $\overline{\mathrm{NND}} = n^{-1} \sum_i d_i$ and $\sigma_{\mathrm{NND}} = \sqrt{n^{-1} \sum_i (d_i - \overline{\mathrm{NND}})^2}$.

Clark–Evans index with Donnelly edge correction. The Clark–Evans index compares the observed mean NND to its expectation under complete spatial randomness [16]:

$$R = \frac{\overline{\mathrm{NND}}}{E[\bar{r}]}, \tag{2}$$

where $R < 1$ indicates clustering and $R > 1$ indicates dispersion. For small, bounded clusters the uncorrected expectation $E[\bar{r}] = 0.5\sqrt{A/n}$ is negatively biased by boundary effects. Following Donnelly [17], we use the corrected expectation

$$E[\bar{r}] = 0.5\sqrt{\frac{A}{n}} + \left(0.0514 + \frac{0.041}{\sqrt{n}}\right)\frac{P}{n}, \tag{3}$$

where A is the convex-hull area of the cluster and P is its perimeter.

Fractal dimension. Boundary complexity is quantified by the box-counting fractal dimension:

$$D = -\lim_{\epsilon \to 0} \frac{\log N(\epsilon)}{\log \epsilon}, \tag{4}$$

where $N(\epsilon)$ is the number of boxes of side length ϵ that intersect the cluster's convex hull. In practice D is estimated as the slope of $\log N(\epsilon)$ versus $\log(1/\epsilon)$ over a range of scales expressed as fractions of the hull's bounding-box diagonal. Because the metric is computed on the convex hull rather than the raw point set, it measures the area-filling complexity of the hull boundary; for the small clusters that dominate the dataset (median eight points) it is correspondingly coarse, and we treat it as a descriptive shape index rather than a true fractal estimate.

Compactness (form factor).

$$C = \frac{4\pi A}{P^2}, \tag{5}$$

where A and P are the convex-hull area and perimeter. $C = 1$ for a circle and decreases as the shape becomes more elongated or irregular.

Aspect ratio. The aspect ratio is the ratio of the major to minor axis of the minimum-area bounding rectangle:

$$\mathrm{AR} = \frac{\ell_{\max}}{\ell_{\min}}. \tag{6}$$

Alpha compactness.

$$\mathrm{AC} = \frac{\mathrm{area}(\mathcal{A}_\alpha)}{\mathrm{area}(\mathcal{H})}, \tag{7}$$

where $\mathcal{A}_\alpha$ is the alpha shape of the point set and $\mathcal{H}$ is its convex hull. Values near 1 indicate dense, convex clusters; lower values indicate concave or elongated clusters with interior voids.

Core-to-edge ratio. The ratio of the convex hull's interior area to its total area, where the interior is the hull eroded inward by a fixed 1 m buffer: $\mathrm{area}(\mathcal{H} \ominus 1\,\mathrm{m})/\mathrm{area}(\mathcal{H})$. Because the erosion width is fixed while clusters vary in size, this ratio is partly a proxy for absolute cluster size (small clusters lose proportionally more area), which should be borne in mind when interpreting it.

Orientation. PCA is applied to the xy coordinates of cluster points; the orientation angle is

$$\theta = \arctan\left(\frac{v_y}{v_x}\right), \tag{8}$$

where $\mathbf{v} = (v_x, v_y)$ is the first principal eigenvector. θ is folded into $[0°, 180°)$ for use as axial data in orientation analysis.

3.4 Morphological Gradient and Candidate Partition

K-means clustering was applied to the standardised feature vector (fractal_dim, clark_evans, aspect_ratio) using $k \in \{2, \dots, 7\}$. The optimal k was selected by jointly evaluating the silhouette coefficient, Calinski–Harabasz (CH) score, and Davies–Bouldin (DB) index. At $k = 2$ these were silhouette $= 0.401$, CH $= 8767$, DB $= 1.073$—the best combined performance across the tested range.

To assess whether the observed silhouette score reflects genuine structure rather than the covariance of correlated marginals, we compared it against a *covariance-preserving* null: 999 surrogate datasets were drawn from a single multivariate Gaussian whose mean and full covariance matrix match the standardised feature matrix, and $k = 2$ K-means was refitted to each. This is a stricter null than independent column-shuffling (which destroys all inter-feature correlation and is therefore beaten by any correlated data); it asks specifically whether the data are *more clustered than a single elongated Gaussian cloud*. The observed silhouette (0.401) exceeded the null mean (0.321) and 95th percentile (0.324), giving $p = 0.001$.

Because a significant silhouette does not by itself establish that two discrete classes exist, we additionally tested for discreteness. Gaussian-mixture model selection by BIC over $k \in \{1, \dots, 7\}$ and the modality of the first principal component (which captures 60% of feature variance) jointly indicate that the morphological features form a continuous gradient rather than two separated modes (Section 4.2). We therefore treat the $k = 2$ partition as an interpretive summary of the dominant compactness–elongation axis, not as evidence of a natural class boundary.

3.5 Orientation Analysis

Cluster orientations $\theta \in [0°, 180°)$ are axial data and must be doubled before applying circular statistics [19]. The Rayleigh test for uniformity was applied to the doubled angles $2\theta \in [0°, 360°)$ using pycircstat. The mean resultant length $r = \| \sum \exp(i \cdot 2\theta_j) \| / n$ and its associated p-value quantify whether orientations depart from a uniform (no preferred direction) distribution. Results were stratified by type.

3.6 Spatial Autocorrelation

Global Moran's I was computed for four cluster-level metrics (fractal_dim, clark_evans, aspect_ratio, compactness) at five spatial thresholds (200, 500, 1000, 2000, 5000 m) using a stratified random subsample of $n =$ 2,000 clusters. A distance-band contiguity at threshold δ ($d_{ij} \le \delta$) was used, with the weights row-standardised ($\sum_j w_{ij} = 1$). With 20 simultaneous tests we applied a Bonferroni-corrected significance threshold $\alpha^* = 0.05/20 = 0.0025$; significance is assessed with the analytical (normal-approximation) p-value, since a 999-permutation p_{sim} floor of 10^{-3} is the coarsest resolution that can reach α^* and a 199-permutation test (floor 5×10^{-3}) cannot. To test whether any autocorrelation is an artefact of upstream detection error rather than ecology, we additionally propagated a 15% random false-negative detection perturbation through the metric computation over 100 realizations and recorded the fraction preserving the sign and Bonferroni significance of I at each metric $\times$ threshold (a detection-robustness gate).

Local Moran's I (LISA; Anselin 18) was computed to identify statistically significant spatial clusters (high–high and low–low) and spatial outliers in fractal dimension.

3.7 Landscape Connectivity

A proximity graph was constructed by connecting all cluster centroids within 200 m [21]. The reach of a cluster is defined as the number of other clusters reachable within the connected component to which it belongs. Betweenness centrality was computed to identify bridge clusters whose removal would fragment the network. Reach distributions were compared between types using the Mann–Whitney U test.

3.8 Held-Out Classifier

To test whether the $k = 2$ partition is recoverable in a feature space not used to define it, we trained a Random Forest classifier using only features *not* input to K-means: compactness, alpha_compactness, core_to_edge_ratio, nnd_mean, nnd_std, orientation, and reach_200m. (We note that `orientation` is axial data on $[0°, 180°)$ entered here as a linear feature, so the tree splits cannot exploit its circular structure; given its low importance below, this does not affect the conclusions.) Spatial block cross-validation (GroupKFold, 5 folds, 100×100 km grid, 12 blocks, mean 1215 clusters/block) was used to prevent spatial leakage. We caution that the blocks are size-imbalanced—the five largest hold roughly four-fifths of the clusters—so the tight fold-to-fold AUC range understates true spatial variability. SHAP values [22] were computed to assess feature importance.

3.9 Stability Gates

To test whether the $k = 2$ partition is a genuine structure or an artefact of the pipeline's free parameters, we re-ran the entire pipeline from the raw points under four gates.

(G1) Scale stability. We re-clustered the raw points at eps $\in \{10, 15, 20, 25, 30, 40\}$ m and, at each scale, recomputed all shape metrics, re-applied the filters, and re-fitted $k = 2$ K-means. A genuine class should be eps-invariant in prevalence and in its feature-space centroid; an artefact of detection chaining rescales with eps.

(G2) Aspect-ratio filter, from raw. The AR ≤ 10 filter is applied to the very axis on which K-means clusters, so it cannot be tested on the already-filtered table. We instead recomputed metrics on all clusters at eps = 20 m with *no* AR cap, then applied caps $\in \{5, 7, 10, 15, 20, \infty\}$, re-fitting K-means at each and tracking type prevalence, the within-Type 0 AR distribution, and the adjusted Rand index (ARI) of membership against the AR ≤ 10 partition.

(G3) Directional second-order structure. Hull aspect ratio is a first-order summary and cannot distinguish a linear (wind-swath) point arrangement from an elongated but space-filling patch. For each cluster we computed the point-scatter linearity $\ell = \lambda_1/(\lambda_1 + \lambda_2)$ from the PCA eigenvalues and a pairwise-displacement axial resultant, and compared Type 0 against Type 1 *within matched hull-AR bins.*

(G4) Detection confound. The upstream detector provides a per-detection confidence (`max_value`) not otherwise used. We tested whether per-cluster detection covariates (mean/SD confidence, n, density) predict type (5-fold logistic AUC), and whether modest false-negative perturbation (dropping 10–20% of detections per cluster and re-fitting) changes type assignments.

4 Results

4.1 Cluster Characterisation

The 14,582 filtered clusters span a broad range of sizes and morphologies (Figure 1). The median cluster contains 8 trees (mean = 13.75, SD = 44.03, max = 2,965), reflecting a highly right-skewed size distribution dominated by small clusters but with a heavy tail of large events. Median area is 388 m^2 (mean = 1,061 m^2, SD = 5,003 m^2). Shapes are moderately compact (mean compactness = 0.595, SD = 0.143, median = 0.609) and moderately elongated (mean AR = 2.548, SD = 1.357, median = 2.211). The Donnelly-corrected Clark–Evans index (mean = 1.766, SD = 0.393, median = 1.753) is greater than unity, nominally indicating more-regular-than-Poisson spacing. We read this cautiously: at a median of eight points per cluster the index is high-variance, and DBSCAN's minimum-density requirement (min_samples within eps) imposes a spacing floor that mechanically inflates apparent regularity, so the value should not be over-interpreted as an ecological property of tree placement. Mean within-cluster NND is 7.79 m (SD = 2.43 m), consistent with individual-tree-scale spacing. Fractal dimension averages 1.650 (SD = 0.077).

Correlation analysis (Figure 2) shows strong positive covariation among size metrics (n, area) and expected trade-offs between compactness and aspect ratio. Figure 3 presents key pairwise scatter plots.

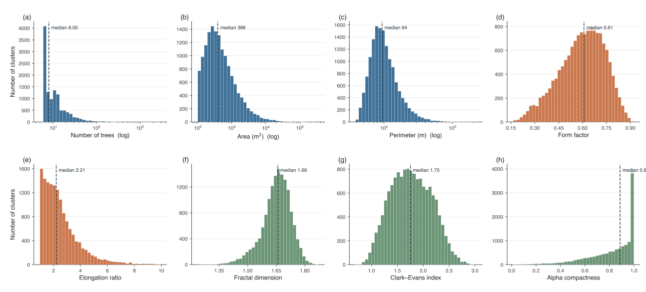

Figure 1: Distributions of key cluster metrics across all 14,582 clusters, coloured by metric family (blue: size/extent; orange: shape/form; green: internal structure). Panels (a)–(c) span two to three orders of magnitude and use a logarithmic x-axis with log-spaced bins; on this scale area and perimeter are approximately log-normal. The remaining panels use linear axes. Dashed lines mark per-metric medians.

The regional spatial distribution of all 14,582 cluster centroids is shown in Figure 4.

4.2 Morphological Gradient and Candidate Partition

K-sweep and selection. Among $k = 2, \ldots, 7$, $k = 2$ gave the best combined validation performance (silhouette = 0.401, CH = 8767, DB = 1.073).

Significance against a covariance-preserving null. The observed silhouette (0.401) exceeded the mean (0.321) and 95th percentile (0.324) of a 999-replicate null drawn from a single multivariate Gaussian matched to the observed feature covariance, yielding $p = 0.001$

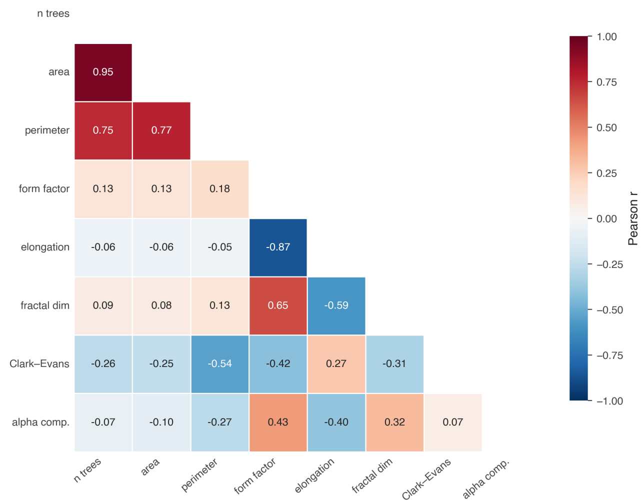

Figure 2: Correlation matrix for cluster-level spatial metrics.

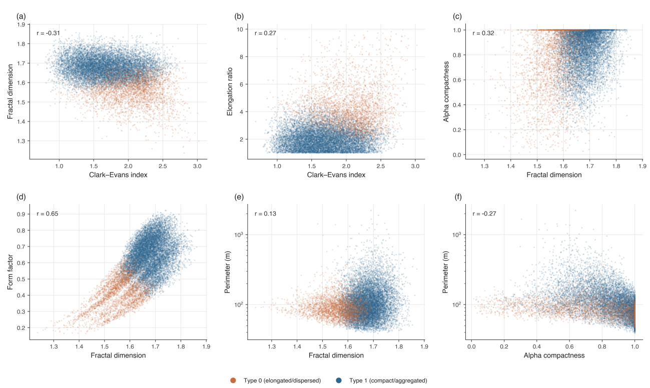

Figure 3: Pairwise scatter plots of selected spatial metrics, coloured by morphological type.

(Figure 5). Because this null preserves all inter-feature correlation, the result indicates that the data are more structured than one elongated correlated cloud—a substantially stronger statement than rejecting the column-shuffled null, which any correlated dataset trivially beats.

Continuum, not dichotomy. Gaussian-mixture BIC decreases monotonically beyond $k = 2$ (BIC: 1.163×10^5, 1.100×10^5, 1.095×10^5, 1.085×10^5, 1.074×10^5, 1.073×10^5 for $k = 1, \ldots, 6$), so no two-component model is privileged; and the first principal component (60% of variance) is unimodal (Sarle bimodality coefficient 0.486, below the 0.555 uniform threshold). Only 2.3% of clusters lie within 5% of the $k = 2$ decision boundary, but this assignment cleanliness is expected when partitioning an elongated unimodal cloud and does not imply two natural modes. We therefore interpret the $k = 2$ partition as two endpoints of a continuous compactness–elongation gradient.

Endpoint descriptions. Table 1 summarises per-endpoint metric means and standard deviations. The *elongated/dispersed* endpoint (*Type 0*; 3,422 clusters, 23.5%) is characterised by high aspect ratio (4.291±1.490), low compactness (0.410±0.093), and elevated Clark–Evans index (2.044±0.360). The *compact/aggregated* endpoint (*Type 1*; 11,160 clusters, 76.5%) shows rounder, denser shapes (compactness = 0.651±0.101, AR = 2.014±0.714) with higher fractal dimension (1.679±0.051) and larger mean area (1,269 m^2 vs. 382 m^2). We use *Type 0* / *Type 1* purely as labels for the two ends of the gradient and attach no disturbance-agent meaning to them here.

The spatial distribution of the two gradient endpoints within the study region is shown in Figure 6; the inset locates the region within Finland.

Size-frequency distributions (E1). For both endpoints a lognormal cluster-size distribution is preferred over a power law (likelihood-ratio test $p < 0.001$; Clauset et al. 23). We therefore do not interpret the nominal power-law exponents as process signatures—once the power law is rejected, its exponent is not a meaningful mechanistic parameter. Descriptively, Type 1 has the heavier upper tail (Figure 7), but this follows directly from its larger mean cluster area (1,269 vs. 382 m^2) and does not by itself imply a contagious generating process.

Position in shape space (E5). Figure 8 plots each cluster in (compactness, aspect_ratio) space against a set of *illustrative* morphological regions loosely informed by qualitative descriptions of wind, beetle, and fire patches in the literature [9; 12; 6]. We stress that these regions are author-defined bounding boxes, not published (compactness, aspect_ratio) envelopes—the cited studies do not report cluster-level shape coordinates by agent—and that any apparent separation is partly circular: aspect ratio is a clustering input and compactness is geometrically coupled to it. The descriptive overlaps (Table 2) therefore restate the gradient definition rather than provide independent ecological validation, and we report them only to locate the two endpoints in a shape space comparable to qualitative literature accounts. Establishing genuine agent correspondence requires overlay with an independent disturbance-agent map (see Discussion).

4.3 Orientation Analysis

Rayleigh tests on doubled orientation angles show no meaningful preferred direction in either type (Figure 9). For Type 0: $r = 0.009$, mean direction = 156.4°, $p = 0.747$. For Type 1: $r = 0.040$, mean direction = 147.0°, $p = 2.0 \times 10^{-8}$. Although the Type 1 result is statistically significant given the large sample ($n = 11{,}160$), the mean resultant length $r \approx 0.04$ is ecologically trivial: orientations are effectively uniformly distributed in both types. The null orientation finding for Type 0 is discussed in Section 5.

Table 1: Per-type means ± SD for all morphological features. Clustering was performed on fractal_dim, clark_evans, and aspect_ratio only; the remaining features are descriptive.

Feature	**Type 0 (elongated/dispersed)** ($n = 3{,}422$)	**Type 1 (compact/aggregated)** ($n = 11{,}160$)
Fractal dimension	1.555 ± 0.069	1.679 ± 0.051
Clark–Evans index	2.044 ± 0.360	1.681 ± 0.363
Aspect ratio	4.291 ± 1.490	2.014 ± 0.714
Compactness	0.410 ± 0.093	0.651 ± 0.101
Alpha compactness	0.713 ± 0.250	0.855 ± 0.161
Core-to-edge ratio	0.662 ± 0.104	0.799 ± 0.089
Mean NND (m)	8.555 ± 2.489	7.561 ± 2.360
Mean area (m^2)	382	1,269
Mean n (trees)	7.7	15.6

Table 2: Position of each endpoint relative to illustrative, author-defined shape regions (E5). These regions are not published reference envelopes and the overlaps are partly circular (see text); they are descriptive, not validating.

Illustrative region	**Type 0**	**Type 1**
"Elongated" (wind-like)	57.0%	1.2%
"Intermediate" (beetle-like)	75.0%	66.4%
"Compact" (fire-like)	32.3%	94.7%

4.4 Spatial Autocorrelation

Figure 10 presents the multi-scale Moran's I heatmap (E6) for four metrics at five distance thresholds. The autocorrelation that is present is weak in magnitude ($I \leq 0.11$ throughout) and, where positive, strongest at the 200–500 m scale and attenuating beyond 2 km. We are deliberately conservative about its significance and its robustness, for two reasons. First, under the analytical p-value only Clark–Evans clears the Bonferroni threshold, and only at the 500 m–5 km scales; the fractal-dimension, aspect-ratio, and compactness autocorrelations are not Bonferroni-significant. Second, the detection-robustness gate shows that the *sign* of I survives a 15% false-negative detection perturbation for Clark–Evans, aspect ratio, and compactness ($\geq$ 96% of realizations at 200–500 m) but *not* for fractal dimension (sign preserved in only 60–72%), whose near-zero autocorrelation is indistinguishable from noise once detection error is propagated. The defensible statement is therefore narrow: within-cluster dispersion (Clark–Evans) shows weak but robust positive spatial autocorrelation at the 0.5–5 km scale, while the shape metrics show at most marginal, detection-sensitive structure. LISA maps (Figure 11) illustrate the spatial structure of fractal-dimension autocorrelation within the study-region display window; labels are computed on the full survey dataset to preserve correct neighbourhood weights, and given the weak global signal for this metric they are best read qualitatively.

4.5 Landscape Connectivity

At the 200 m threshold, the proximity graph over all clusters contains 13,351 edges; the largest connected component spans only 175 of 14,582 clusters and roughly a third of clusters are isolated singletons, so the graph is sparse. Figure 12 shows connectivity reach and bridge clusters within the study-region display window (statistics from the full graph). Type 1 clusters are nominally more connected than Type 0 (median reach = 3 vs. 2; mean reach 11.03 vs. 8.98; Mann–Whitney $U = 17{,}752{,}378$, $p = 1.7 \times 10^{-10}$), but the *effect size is negligible*: the rank-biserial correlation is only $r = 0.07$, the medians differ by a single component, and 34.9% of clusters are isolated singletons. The vanishingly small p reflects the large n, not a meaningful separation, and the difference is in any case a direct consequence of Type 1's larger area and denser packing (larger, denser clusters are mechanically more likely to fall within 200 m of a neighbour). We therefore treat reach as a density correlate, not an independent line of evidence. Because the graph is untimed and unweighted, we do not interpret high-betweenness clusters as outbreak corridors or sentinels; they are geometric articulation points whose ecological relevance is untested. A meaningful connectivity analysis would require a graph whose length scale is matched to a dispersal process—for a bark-beetle hypothesis, a host-conditioned, dispersal-weighted graph at the ∼250 m–1 km scale of *Ips typographus* flight; for wind, storm-front co-occurrence at the tens-of-kilometre scale of a storm footprint—and these are mutually incompatible graphs, which is itself why a single undirected 200 m graph cannot adjudicate either process.

4.6 Feature-Space Coherence Check

The held-out Random Forest classifier achieved AUC = 0.976 ± 0.002 (range 0.973–0.978 across 5 spatial folds; Figure 13). We caution against reading this as independent validation. The "held-out" features are geometrically coupled to the K-means inputs—compactness, for example, correlates with aspect ratio at $r = -0.865$ and alone attains a univariate AUC of about 0.96—so a high AUC is largely a restatement of the K-means boundary through correlated geometry rather than corroboration from new information. We report it only as an internal coherence check; on an imbalanced 76.5/23.5 partition, ROC-AUC should in future be complemented by precision–recall AUC and per-class recall.

SHAP mean absolute values ranked features as follows: compactness (0.226) > core_to_edge_ratio (0.068) > nnd_mean (0.037) > orientation (0.034) > alpha_compactness (0.019) > nnd_std (0.010) > reach_200m (0.005). Compactness dominates because it is geometrically correlated with aspect ratio (a K-means input). Crucially, reach_200m—the only feature in this set that is independent of cluster shape—ranks last, confirming that high AUC is primarily driven by correlated shape dimensions rather than landscape connectivity.

This same asymmetry is what licenses reporting the gradient while withholding the discrete typology. The detector confound (G4) predicts the *discrete* type at AUC = 0.727, yet a cluster's position along the *continuous* gradient (first principal component of the three features) correlates with detection confidence at only $\rho = 0.07$. The discrete labels sit close to the detector's decision surface; the continuous morphospace coordinate does not. The gradient is therefore the level at which the signal is least entangled with the instrument, and the typology is the level at which it is most entangled—hence the former is reported and the latter withheld.

4.7 Stability Gates

The four gates jointly indicate that the $k = 2$ split is not a discrete typology but a convenience cut through a continuous gradient, and that one of its endpoint labels is unsupported.

(G1) The partition is not scale-stable. Re-clustering from raw across eps $\in \{10, \ldots, 40\}$ m, the Type 0 fraction slides monotonically (28.6% at 10 m, 25.8%, 23.5% at 20 m, 24.0%, 23.4%, 22.9% at 40 m) and the Type 0 centroid drifts continuously rather than holding position (aspect ratio $2.97 \rightarrow 4.29 \rightarrow 4.79$; compactness $0.544 \rightarrow 0.410 \rightarrow 0.372$ across eps 10/20/40 m). A genuine class would be eps-invariant; this rescaling with the clustering radius is the signature of detection chaining, not a fixed disturbance class. BIC favours more than two components at every eps.

(G2) The headline prevalence is an artefact of the AR cap. Re-clustering from raw at eps = 20 m with no AR cap (14,644 clusters) and sweeping the cap shows the Type 0 fraction is conditional on where the cap falls: 33.1% (AR $\leq$ 5), 27.0% ($\leq$ 7), 23.5% ($\leq$ 10, the published value), 21.8% ($\leq$ 15), down to 21.3% (uncapped). The within-Type 0 AR distribution is pressed against the cap (maximum 9.94 at AR $\leq$ 10) and, once released, extends as a single smooth tail to AR $\approx$ 30 with no second mode. Membership ARI against the AR $\leq$ 10 partition decays monotonically as the tail is restored ($1.00 \rightarrow 0.92 \rightarrow 0.89$). Aspect ratio is therefore a continuum, and the AR $\leq$ 10 cap manufactured the clean "23.5%" figure.

(G3) No directional signal beyond hull shape. Within matched hull-AR bins, Type 0 and Type 1 have statistically indistinguishable point-scatter linearity (Figure 14; e.g. AR $\in (2, 2.5]$: 0.829 vs. 0.830, $p = 0.86$; AR $\in (3, 4]$: 0.916 vs. 0.915, $p = 0.73$; Type 1 marginally higher in the $(2.5, 3]$ bin). The overall impression that Type 0 is "more linear" (Figure 14a) is entirely a hull-AR effect; once hull elongation is controlled (Figure 14b) the elongated endpoint carries no extra line-like (wind-swath) structure. Together with the orientation null (Section 4.3), this empirically removes the basis for a "wind-thrown" interpretation.

(G4) A measurable detector confound. Per-cluster detection-confidence covariates predict type at AUC = 0.727 (5-fold), and dropping 10–20% of detections per cluster flips the type of 5.9–7.5% of clusters (ARI 0.76–0.70). The typology is thus partly entangled with detector behaviour; cleanly separating the two requires stratified detection precision/recall against ground truth, which the present data lack.

5 Discussion

A morphological gradient, not a disturbance-agent typology. It is tempting to map the two endpoints onto dominant Finnish disturbance agents—the elongated, internally dispersed Type 0 onto wind-felled strips [6; 7], and the compact, denser Type 1 onto radially spreading beetle kill [4; 12; 13]. We deliberately resist this mapping, because the evidence that would license it is either absent or circular. The shape-space "reference zones" are author-defined boxes in a space whose axes (aspect ratio, and the compactness coupled to it) *define* the gradient, so their overlap with the endpoints is a restatement of the clustering, not external corroboration. Several confounders independently block attribution: the dataset carries no species information, so a beetle-killed spruce cannot be distinguished from a drought- or pathogen-killed pine, nor a compact patch from fire, snow-load, or small salvage/clearcut polygons; the `alku_aika` field records flight-strip acquisition time, not mortality date, severing any link to specific storms or outbreak years; and the single 2023 snapshot cannot separate a recent event from a years-old palimpsest of snags. A compact morphology, in particular, is the default shape of almost any small mortality patch and is therefore the least diagnostic possible signal—making a 76.5% "compact" prevalence uninformative about agent identity.

Three of our own results point the same way: this is geometry, not agent. The most telling evidence is internal and convergent. (i) *Orientation.* The elongated endpoint shows no preferred orientation ($r = 0.009$, $p = 0.747$)—yet directional alignment with storm tracks is the single most diagnostic fingerprint of windthrow. An elongated class with no net direction is precisely what one would *not* expect from wind, and it is the main reason we decline the "wind-thrown" label. (ii) *Scale dependence.* The prevalence of the elongated endpoint moves with the DBSCAN radius (22.9% at 40 m to 28.6% at 10 m); a genuine ecological class should be eps-stable, whereas a chaining artefact rescales with eps. (iii) *Connectivity.* The only shape-orthogonal feature, 200 m reach, carries essentially no discriminative signal (SHAP 0.005, last of seven). Read together—nondirectional elongation, eps-dependent prevalence, and no independent connectivity signal—the most parsimonious account is that the typology summarises a continuum of *detection-and-clustering geometry* (how dead- tree detections chain together at a 20 m radius, with mean within-cluster spacing of only 7.79 m), rather than two distinct disturbance processes. The earlier ecological reading of the size-frequency tails is correspondingly withdrawn: with the power law rejected in favour of a lognormal for both endpoints, the heavier Type 1 tail is explained by its larger mean area, not by contagious spread.

Interpretation of the held-out classifier AUC. The AUC = 0.976 shows that the two endpoints occupy largely non-overlapping regions of the held-out feature space, but the SHAP decomposition shows why this is not independent validation: performance is driven almost entirely by compactness and core-to-edge ratio, both geometrically correlated with aspect ratio (a K-means input). The weakest predictor, reach_200m (SHAP = 0.005), is the only feature capturing landscape connectivity independently of cluster shape; its near-zero importance means the high AUC restates the K-means boundary through correlated geometry rather than confirming it from new information. The honest reading is the negative one: on the single axis genuinely independent of shape, the two endpoints are not distinguishable.

Limitations. Beyond the attribution caveats above, several additional limitations merit note. The upstream deep-learning detection model is reported here without validation metrics (precision, recall), and gate G4 shows its confidence score alone predicts type at AUC = 0.727—so detection error, likely structured by forest type, canopy density, and flight-strip geometry, is partly entangled with the morphology we measure. Fully separating the two requires stratified detection precision/recall against ground truth, which the present data lack; this is the single most important outstanding check, and the Clark–Evans index, computed on a median of eight points per cluster, is especially sensitive to it. The study area is regional (southeastern Finland), not national, so reported prevalences are not national statistics. Gate G1 confirms that the eps = 20 m clustering scale partly determines the elongation it then measures. Finally, because standing dead trees persist for years in boreal stands, a single-year snapshot cannot date mortality: what we treat as one "cluster" may aggregate trees killed by different agents in different years that happen to be co-located, so the unit of analysis is itself only loosely defined—a fundamental limit on any morphology-to-process inference from one acquisition.

Future work. The decisive next step is external validation that breaks the circularity of a shape-only study. Overlaying a species layer from Finland's multi-source national forest inventory and an independent disturbance-agent map would convert the morphological endpoints from an unsupervised gradient into a testable, supervised agent classification, and would let us check whether the compact endpoint is genuinely beetle-dominated or a mix of fire, drought, snow-load, and salvage. We note one concrete obstacle for the wind-versus-beetle question specifically: the publicly available pan-European attribution products [11] group wind and bark beetle into a single combined class, so they cannot by themselves separate the two agents our endpoints might be hypothesised to represent—a dedicated regional agent layer (e.g. Metsäkeskus damage records) would be required. Multi-temporal imagery—feasible from the same pipeline for prior years—would add the missing dimension: it would date mortality, permit a true wave-front and storm-track alignment test for the elongated endpoint, and turn the connectivity graph from a static density correlate into a flow model that asks whether this year's morphology predicts next year's mortality. The pipeline itself is computationally scalable and, once validated, could support comparative analysis across the Nordic countries.

6 Conclusion

We have presented a regional-scale morphological analysis of 14,582 dead-tree clusters identified from a 2023 aerial mortality survey in southeastern Finland. Using DBSCAN clustering, Donnelly-corrected Clark–Evans statistics, and a suite of complementary shape metrics, we show that the morphology of detected mortality clusters is organised by a dominant compactness–elongation gradient. This gradient is not a statistical artefact of correlated features: the $k = 2$ silhouette (0.401) exceeds a covariance-preserving null at $p = 0.001$. The *discrete* two-class reading of that gradient, however, does not survive scrutiny. Mixture-model selection and the unimodality of the dominant axis already indicate a continuum rather than two modes; four stability gates confirm it. Re-clustering from raw, the partition is not invariant to the DBSCAN radius (G1), its 23.5%/76.5% split is an artefact of the aspect-ratio filter and dissolves into a single continuous tail once that filter is released (G2), the elongated class carries no directional structure beyond hull shape (G3, consistent with the orientation null), and it is partly predictable from detector confidence alone

(G4). The held-out Random Forest (AUC = 0.976) reproduces the split only through geometric redundancy among shape descriptors—the one shape-orthogonal feature, landscape connectivity, adds no signal.

The framework demonstrates that remote-sensing-derived point clouds of tree mortality contain a recoverable and reproducible morphological gradient—but that the temptation to discretise it into disturbance types must be resisted on the present evidence. The discrete partition is an artefact of clustering scale and a filter applied to its own defining axis, with no second-order content and a measurable detector confound; the gradient is what is real. Rather than a ready "first filter" for the field, the pipeline is best positioned as a hypothesis-generating characterisation whose ecological meaning must be established by overlay with independent disturbance-agent maps, species layers, and multi-temporal imagery—a programme we outline as the natural next step.

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Figure 4: Map of dead-tree cluster centroids (n = 14,582) across the southeastern Finland study region (EPSG:3067).

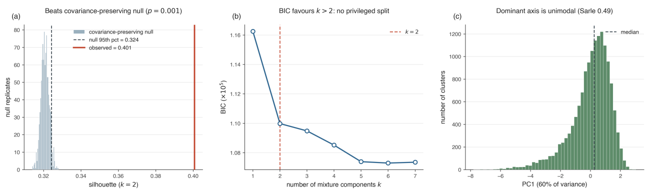


Figure 5: Significance and discreteness of the $k = 2$ partition. **(a)** The observed silhouette (red line, 0.401) exceeds a 999-replicate *covariance-preserving* null (single multivariate Gaussian matched to the feature covariance; mean 0.321, 95th pct 0.324), $p = 0.001$—a far stricter test than column-shuffling. **(b)** Gaussian-mixture BIC decreases monotonically past $k = 2$, so no two-component model is privileged. **(c)** The first principal component (60% of variance) is unimodal (Sarle coefficient 0.486). Together: real structure, but a continuous gradient rather than two discrete classes.

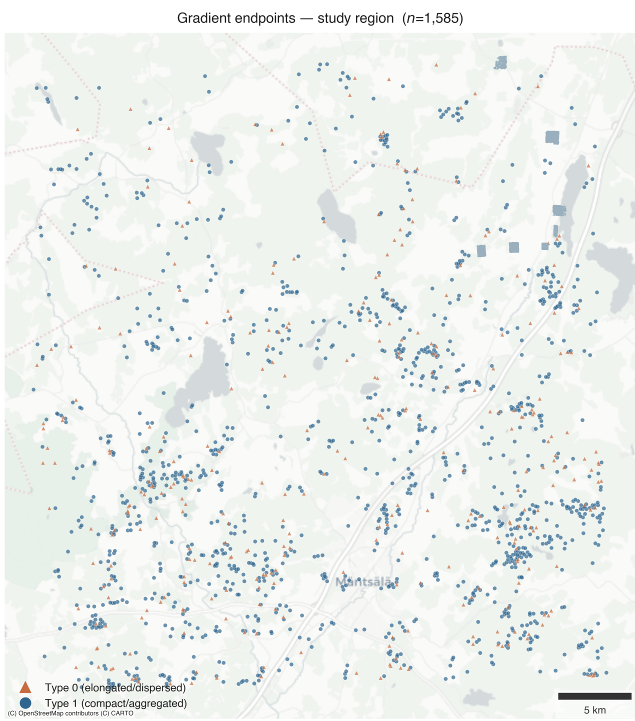


Figure 6: Spatial distribution of the two gradient endpoints within the study region. Blue circles: Type 1 (compact/aggregated); brown triangles: Type 0 (elongated/dispersed). Inset: all 14,582 clusters in the survey area for geographic context.

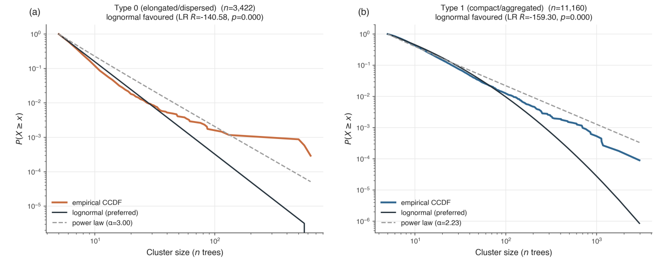


Figure 7: Complementary cumulative distribution functions (CCDF) of cluster size (n) per endpoint. Both are better described by a lognormal than a power law; Type 1 has the heavier upper tail, consistent with its larger mean cluster area.

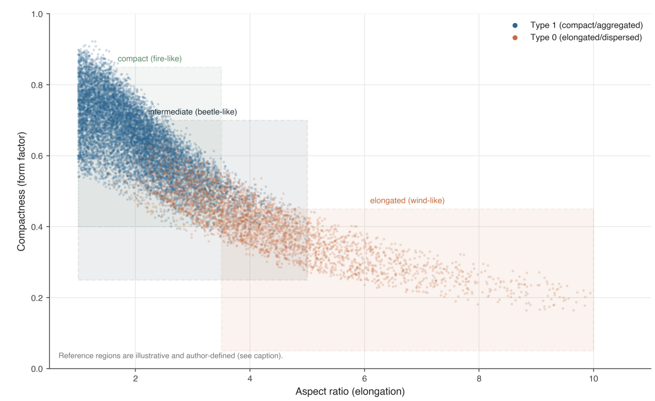


Figure 8: Shape-space scatter (compactness vs. aspect ratio) per endpoint, with *illustrative, author-defined* shape regions overlaid. The separation is partly circular because aspect ratio defines the gradient and compactness is coupled to it; the regions are not published per-agent envelopes.

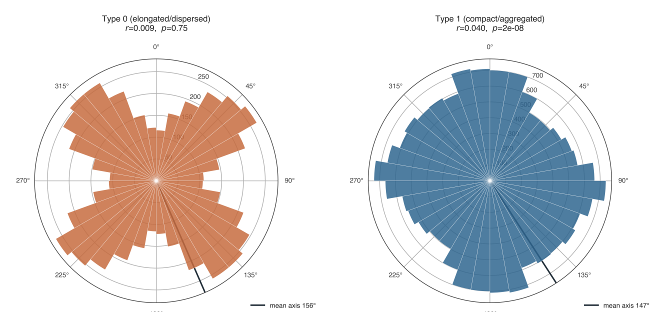


Figure 9: Polar rose plots of cluster orientation per type (axial data, 10° bins). Neither type shows a meaningful preferred direction ($r < 0.05$ in both cases).

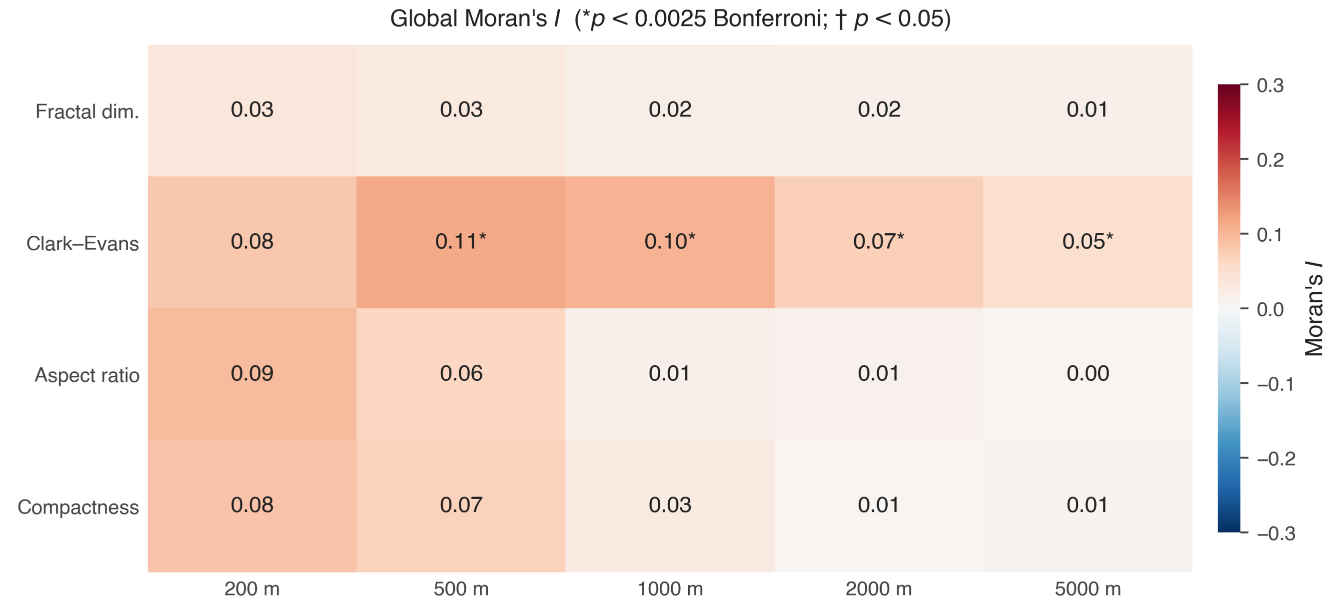


Figure 10: Moran's I heatmap (E6): four metrics (rows) $\times$ five distance thresholds (columns), analytical p. Cells marked $*$ clear the Bonferroni threshold $\alpha^* = 0.0025$, † are nominally significant ($p < 0.05$). Autocorrelation is weak throughout ($I \leq 0.11$) and strongest at 200–500 m; only Clark–Evans is Bonferroni-significant, and a 15% false-negative detection perturbation leaves the sign of I stable for all metrics except fractal dimension.

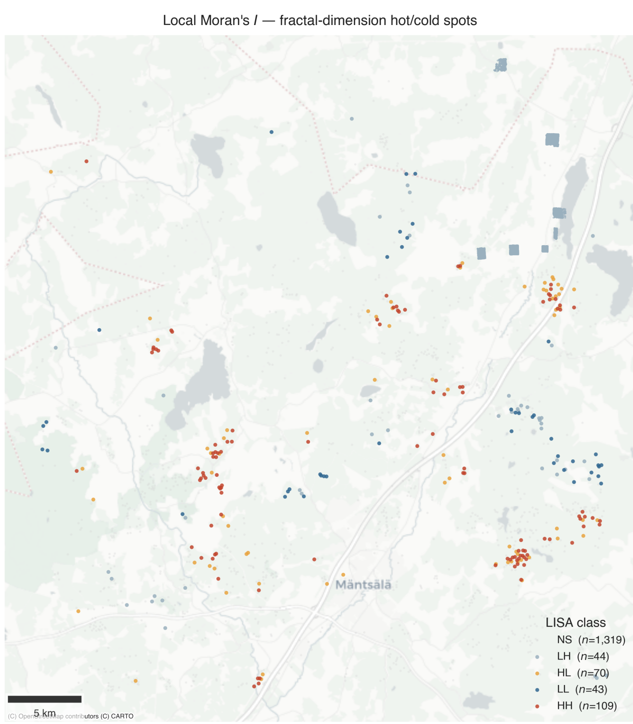
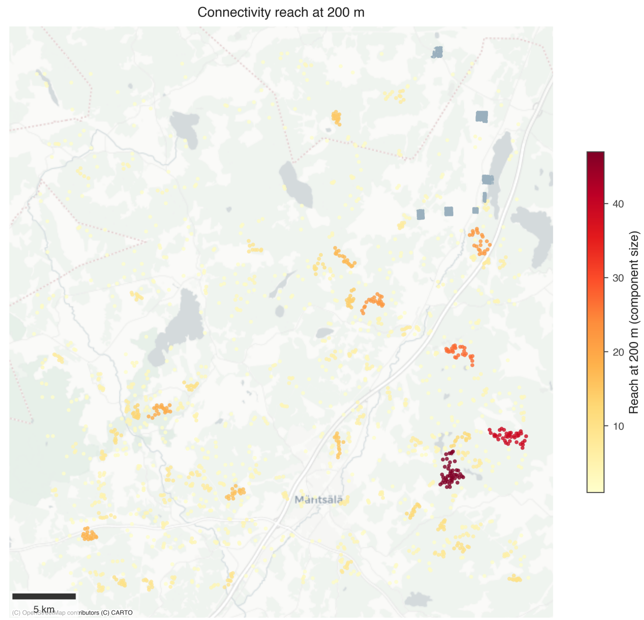
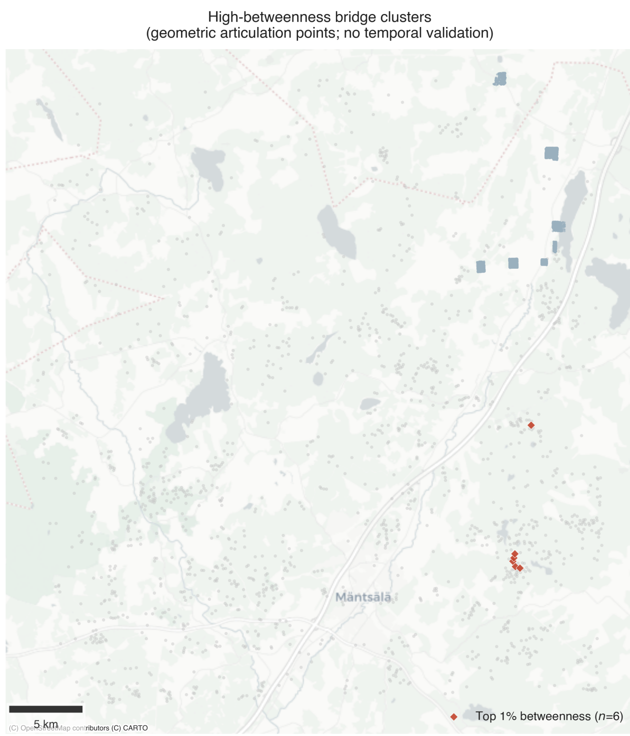


Figure 11: LISA hotspot map for fractal dimension in the study-region display window. Labels computed on the full survey dataset. Red: high–high clusters; blue: low–low clusters. Inset: full-survey cluster distribution for geographic context.



(a) Connectivity reach at 200 m (component size). Inset: full-survey cluster distribution.



(b) Top 1% betweenness clusters (red) as geometric articulation points (no temporal validation). All metrics computed on the full graph.

Figure 12: Landscape connectivity in the study-region display window.

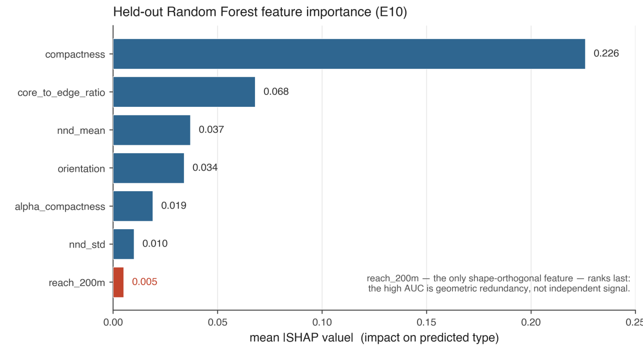


Figure 13: Mean |SHAP| feature importance for the held-out Random Forest (E10). Compactness dominates; reach_200m (the only shape-orthogonal feature, highlighted) ranks last—so the high AUC reflects geometric redundancy among shape descriptors rather than independent signal.

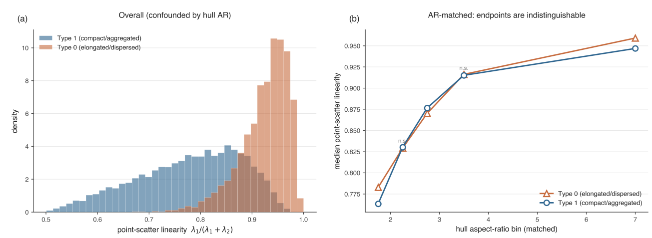


Figure 14: Second-order anisotropy gate (G3). **(a)** Overall, the elongated endpoint appears more linear—but this is confounded by hull aspect ratio. **(b)** Matched on hull aspect ratio, the two endpoints' median point-scatter linearity is indistinguishable (n.s. where shown), so the elongated endpoint carries no directional point-process signal beyond hull shape.

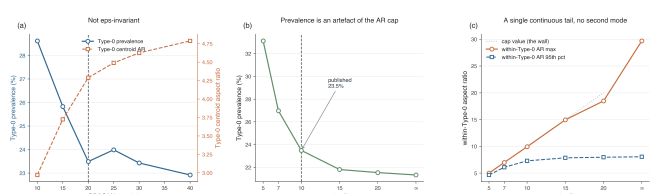


Figure 15: Stability gates. The Type 0 fraction is not invariant to DBSCAN eps (G1) and is conditional on the aspect-ratio cap (G2); the elongated endpoint's AR distribution forms a single continuous tail once the cap is released, with no second mode.